

LISBOA GAGO COUTINHO, Portugal

WMO: 085790

Lat: 38.7662N

Lon: 9.1275W

Elev: 344

StdP: 14.51

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.5	42.6	25.9	20.1	47.5	29.6	24.0	49.0	22.4	57.1	19.8	55.2	6.6	50	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.2	92.7	68.2	89.0	67.3	85.4	66.5	70.7	86.1	69.4	82.7	68.4	80.2	6.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	100.7	71.9	65.7	96.0	71.7	64.6	92.3	71.4	34.7	86.3	33.6	83.0	32.7	80.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.3	14.4	36.3	99.9	2.3	4.2	34.6	102.9	33.2	105.3	31.9	107.6	30.2	110.7		
(5)			32.9	73.8	2.9	1.3	30.8	74.7	29.2	75.5	27.6	76.3	25.5	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	52.3	53.7	57.7	60.3	64.6	70.1	72.9	74.3	71.4	66.0	57.9	53.3		
	DBStd	9.01	4.24	3.78	4.47	5.09	5.57	5.47	4.86	4.74	4.42	4.42	4.39	4.32		
	HDD50	59	26	12	2	0	0	0	0	0	0	0	1	18		
	HDD65	1799	393	317	228	161	74	9	0	0	1	39	215	362		
	CDD50	4779	98	115	242	308	454	603	709	752	642	496	239	121		
	CDD65	1046	0	0	3	19	63	163	245	287	193	70	3	0		
	CDH74	6717	0	0	23	108	439	1112	1710	2037	1031	256	1	0		
	CDH80	2373	0	0	1	13	133	413	646	787	328	52	0	0		
(14) Wind	WSAvg	7.1	6.4	6.9	7.4	7.5	7.6	7.6	8.0	7.5	6.5	6.5	6.3	6.7		
(15) Precipitation	PrecAvg	28.9	4.1	3.6	2.8	2.4	1.9	0.7	0.1	0.2	1.2	3.7	4.8	4.0		
	PrecMax	46.8	13.9	11.7	10.5	6.4	6.1	3.9	1.7	2.3	5.1	10.3	14.7	15.1		
	PrecMin	16.4	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2		
	PrecStd	8.1	3.0	2.6	2.3	1.6	1.6	0.9	0.3	0.4	1.3	2.9	3.4	3.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	67.1	76.8	81.8	89.3	94.8	97.2	97.9	92.1	85.6	72.1	64.6		
		MCWB	56.9	56.1	58.1	62.8	66.7	68.8	70.3	68.8	66.9	63.2	62.1	60.1		
	2%	DB	61.8	63.9	72.8	77.1	83.7	89.1	91.7	92.6	87.8	80.1	68.7	62.9		
		MCWB	57.1	54.7	57.3	61.0	64.4	67.7	68.4	68.6	66.0	63.5	61.8	59.7		
	5%	DB	60.2	61.8	69.3	73.1	79.0	84.7	86.9	88.2	83.9	76.3	66.6	61.7		
		MCWB	56.4	54.6	56.4	59.6	62.9	66.2	67.6	67.5	65.8	63.0	60.8	58.6		
	10%	DB	58.9	60.0	65.5	69.1	74.4	80.4	82.6	83.9	79.8	73.1	64.7	60.2		
		MCWB	55.3	54.4	55.5	58.1	61.5	65.0	66.5	66.5	64.6	62.5	59.7	57.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	59.5	61.4	65.7	69.3	71.1	72.2	72.6	70.3	69.2	66.3	62.3		
		MCDB	61.7	62.0	70.6	77.8	83.8	88.9	90.8	90.0	84.1	73.8	69.0	63.3		
	2%	WB	59.1	58.3	59.4	62.5	66.2	69.3	70.4	70.7	68.9	67.7	64.2	61.0		
		MCDB	60.5	61.1	65.9	74.0	79.2	84.6	86.5	85.2	80.2	72.4	66.7	62.3		
	5%	WB	57.6	56.9	58.3	60.6	64.2	67.7	68.9	69.3	67.7	66.4	62.6	59.3		
		MCDB	59.4	60.1	64.4	70.1	75.4	81.0	83.1	81.9	77.3	71.3	65.3	61.1		
	10%	WB	56.0	55.4	57.2	59.0	62.6	66.0	67.5	68.1	66.6	65.1	60.8	57.5		
		MCDB	58.2	58.7	63.3	66.5	71.9	78.0	80.1	80.0	75.1	70.0	63.6	59.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.7	13.3	13.7	14.7	16.5	17.6	18.2	15.9	12.9	10.6	9.8		
		MCDBR	10.9	14.1	19.7	20.1	22.4	23.5	24.6	24.4	22.1	18.5	12.2	9.8		
		MCWBR	7.4	8.0	9.3	9.8	10.4	10.2	9.6	9.5	8.7	8.0	7.2	7.1		
		MCDBR	8.3	11.0	15.1	17.8	19.0	21.5	21.7	20.3	17.1	12.6	9.6	8.4		
	5% WB	MCWBR	7.2	7.5	8.2	9.1	9.8	9.8	9.2	8.7	7.6	6.8	7.0	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.339	0.368	0.385	0.393	0.397	0.382	0.395	0.394	0.360	0.336	0.327		
	taud		2.517	2.476	2.405	2.351	2.359	2.366	2.401	2.374	2.389	2.503	2.528	2.516		
	Ebn at Noon		270	281	282	283	282	279	283	277	270	270	264	260		
	Edh at Noon		24	29	34	38	39	39	37	37	35	28	24	23		
(44) All-Sky Solar Radiation	RadAvg		711	1028	1396	1796	2155	2329	2372	2153	1684	1158	791	642		
	RadStd		89	120	148	147	144	121	94	82	107	95	97	72		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page